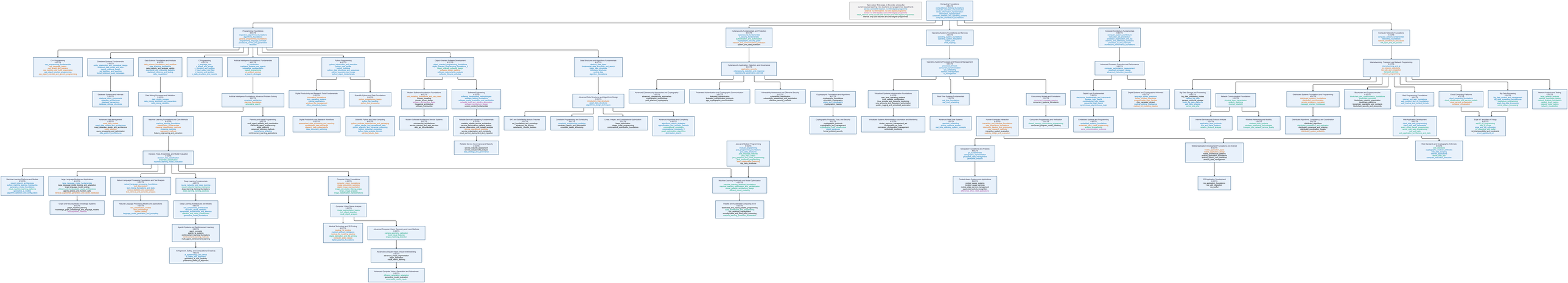

Topic colour: first scope, in this order, among the
current courses teaching it (by teachers' and programmes' department)
service: some DISI teacher, no DISI degree programme
external: no DISI teacher, no DISI degree programme
borrow: no DISI teacher, some DISI degree programme
weak_internal: some (not all) DISI teachers and DISI degree programmes
internal: only DISI teachers and DISI degree programmes
Computing Foundations
6 ECTS
computational_thinking_foundations
computer_operation_and_execution
binary_information_representation
information_representation
computer_software_and_operating_systems
computer_architecture_foundations
Programming Foundations
6 ECTS
imperative_algorithmic_foundations
algorithmic_foundations
general_programming_foundations
programming_language_concepts
procedural_control_and_parameters
recursion
C++ Programming
6 ECTS
cpp_programming_fundamentals
cpp_language_basics
cpp_arrays_and_pointers
cpp_memory_and_build_organization
cpp_object_oriented_programming
cpp_object_oriented_and_generic_programming
Database Systems Fundamentals
5 ECTS
entity_relationship_and_conceptual_design
relational_data_model_and_keys
logical_relational_design
sql_definition_and_querying
formal_relational_query_languages
Database Systems and Internals
4 ECTS
relational_dbms_components
database_architectures
database_transactions
database_storage_structures
Advanced Data Management
6 ECTS
nosql_data_models
nosql_data_models_and_architectures
nosql_database_design_and_architecture
nosql_querying
query_processing_and_optimization
data_platform_architectures
Data Science Foundations and Analysis
6 ECTS
data_science_foundations_and_workflow
data_analytics_foundations
data_integrity_and_analysis_validity
descriptive_and_inferential_statistics
statistical_inference_and_testing
data_visualization
Data Mining Processes and Validation
3 ECTS
data_mining_process
data_mining_workflows_and_preparation
data_mining_validation
Machine Learning Foundations and Core Methods
6 ECTS
machine_learning_foundations
classification_fundamentals
classical_classification_methods
clustering_methods
regression_methods
feature_engineering_and_extraction
Decision Trees, Ensembles, and Model Evaluation
3 ECTS
decision_tree_classification
ensemble_classification
machine_learning_model_evaluation
Machine-Learning Platforms and Models
6 ECTS
neural_network_architectures
python_machine_learning_frameworks
generative_model_foundations
cloud_machine_learning_platforms
generative_ai_models
algorithm_selection_and_configuration
Large Language Models and Applications
6 ECTS
large_language_model_fundamentals
large_language_model_training_and_adaptation
large_language_model_scaling
efficient_and_compressed_neural_architectures
agents_actions_and_function_calls
retrieval_augmented_generation_and_vector_databases
Graph and Neurosymbolic Knowledge Systems
3 ECTS
graph_machine_learning
knowledge_graph_embeddings_and_language_models
neurosymbolic_methods
Natural Language Processing Foundations and Text Analysis
5 ECTS
natural_language_processing_foundations
text_tokenization
text_mining_foundations_and_tools
corpus_statistics_and_association
text_retrieval_and_semantic_analysis
Natural Language Processing Models and Applications
4 ECTS
text_classification_models
word_embeddings
opinion_mining
language_model_generation_and_prompting
Deep Learning Fundamentals
4 ECTS
neural_networks_and_deep_learning
neural_network_training_mechanics
deep_learning_training_foundations
deep_learning_training_practices
Deep Learning Architectures and Models
5 ECTS
cnn_components_architectures
recurrent_neural_networks
transformer_architectures_and_attention
attention_and_vision_transformers
generative_model_foundations
Agentic Systems and Reinforcement Learning
5 ECTS
agent_concepts
agentic_ai_systems
reinforcement_learning_foundations
reinforcement_learning_methods
multi_agent_reinforcement_learning
AI Alignment, Safety, and Computational Creativity
4 ECTS
ai_epistemology_and_ethics
ai_safety_and_alignment
generative_ai_and_creativity
preference_based_ai_alignment
Computer Vision Foundations
6 ECTS
computer_vision_foundations
image_acquisition_sampling
digital_image_representation
image_convolution_filtering_edges
binary_image_analysis
image_classification_representations
Computer Vision Scene Analysis
3 ECTS
image_segmentation_basics
cnn_object_detection
visual_object_analysis
Medical Technology and 3D Printing
6 ECTS
medical_3d_printing
medical_artificial_intelligence
medical_3d_modeling_pipeline
digital_fabrication_and_3d_printing
computer_aided_design
digital_graphics_foundations
Advanced Computer Vision: Geometry and Local Methods
3 ECTS
camera_geometry_calibration
local_visual_features
shape_matching_detection
Advanced Computer Vision: Visual Understanding
3 ECTS
advanced_image_segmentation
depth_estimation
visual_metric_learning
Advanced Computer Vision: Generation and Robustness
3 ECTS
diffusion_generation_adaptation
generative_model_evaluation
adversarial_neural_inputs
Machine-Learning Workloads and Model Optimization
4 ECTS
machine_learning_workload_foundations
machine_learning_optimization_and_parallelization
neural_network_architecture_design
efficient_neural_modeling
Parallel and Accelerated Computing for AI
5 ECTS
distributed_and_hybrid_parallel_programming
cuda_architecture_and_programming
hpc_workload_management
reconfigurable_and_fixed_point_computing
machine_learning_processor_acceleration
C Programming
6 ECTS
c_language_core
c_arrays_and_strings
c_functions_and_building
c_input_output_and_files
c_memory_and_pointers
c_data_structures_and_records
Artificial Intelligence Foundations: Fundamentals
6 ECTS
ai_introduction
intelligent_systems_and_agents
knowledge_representation
logical_programming
ai_problem_solving
ai_search_strategies
Artificial Intelligence Foundations: Advanced Problem Solving
3 ECTS
constraint_satisfaction
planning_foundations
adversarial_search
Planning and Agent Programming
6 ECTS
multi_agent_artifacts_and_coordination
intentional_agent_programming
strips_planning
advanced_planning_methods
maintenance_analytics
reinforcement_learning_applications
Python Programming
6 ECTS
python_introduction_and_execution
python_core_syntax
python_functions
python_data_structures_and_sequences
python_execution_and_scope
python_object_fundamentals
Digital Productivity and Research Tools Fundamentals
5 ECTS
informatics_foundations
linux_operating_systems
internet_applications
windows_file_management
office_productivity_basics
Digital Productivity and Research Workflows
4 ECTS
spreadsheet_data_processing_and_reporting
spreadsheet_data_exchange
research_data_presentations
latex_document_authoring
Scientific Python and Data Foundations
3 ECTS
python_programming_basics
python_file_handling
python_text_processing
Scientific Python and Data Computing
6 ECTS
python_modules_dependencies_and_packaging
python_tabular_and_spreadsheet_data
python_scientific_and_numerical_computing
python_numerical_computing
python_data_analysis_libraries
pandas_tabular_data
Object-Oriented Software Development
6 ECTS
object_oriented_programming_foundations
object_oriented_programming_foundations_2
object_oriented_software_design
software_design_patterns
software_requirements_analysis
software_lifecycle_activities
Modern Software Architecture Foundations
6 ECTS
uml_modeling_foundations_and_use_cases
domain_modeling
domain_driven_design
software_architecture_styles
hexagonal_architecture
software_frameworks
Modern Software Architecture Service Systems
3 ECTS
microservice_architectures
service_oriented_and_web_services
web_api_documentation
Software Engineering
6 ECTS
software_development_methodologies
software_testing_foundations
software_quality_evaluation_and_certification
software_build_and_delivery_automation
software_architecture_design
version_control_fundamentals
Reliable Service Engineering Fundamentals
6 ECTS
scalable_reliable_service_architecture
service_performance_capacity_testing
service_observation_and_outage_analysis
service_operational_response
service_organization_and_complexity
cloud_service_deployment_and_migration
Reliable Service Governance and Maturity
3 ECTS
service_maturity_assessment
service_cost_benefit_analysis
data_strategy_and_governance
Data Structures and Algorithms Fundamentals
6 ECTS
abstract_data_types
fundamental_data_structures_and_search
linear_data_structures
dictionary_structures
sorting_data
algorithm_foundations
Advanced Data Structures and Algorithmic Design
3 ECTS
introductory_tree_structures
graph_data_structures
algorithmic_design_techniques
SAT and Satisfiability Modulo Theories
3 ECTS
sat_foundations_and_encodings
advanced_sat_solving
satisfiability_modulo_theories
Constraint Programming and Scheduling
3 ECTS
constraint_programming_modeling
constraint_search_and_optimization
constraint_based_scheduling
Linear, Integer, and Combinatorial Optimization
3 ECTS
linear_optimization
integer_linear_programming
combinatorial_optimization_foundations
Advanced Algorithms and Complexity
6 ECTS
algorithmic_search_strategies
approximation_and_heuristic_methods
computational_complexity
computational_complexity_2
graph_optimization_algorithms
optimization_search
Java and Modular Programming
9 ECTS
modular_programming
java_programming_foundations
java_data_structures
java_language_features
java_input_output
java_graphical_and_event_programming
java_functional_programming
latex_collaborative_authoring
cpp_data_structures
Cybersecurity Fundamentals and Protection
6 ECTS
cybersecurity_fundamentals
it_security_fundamentals
authentication_and_authorization
cryptographic_security_goals
network_and_communications_protection
system_and_data_protection
Cybersecurity Application, Detection, and Governance
3 ECTS
application_security
cybersecurity_detection_and_response
cybersecurity_governance_and_risk
Advanced Cybersecurity Approaches and Cryptography
3 ECTS
advanced_cybersecurity_approaches
cryptographic_randomness
post_quantum_cryptography
Federated Authentication and Cryptographic Communication
3 ECTS
federated_authentication
federated_authentication_protocols
pgp_cryptographic_communication
Vulnerability Assessment and Offensive Security
3 ECTS
vulnerability_identification
vulnerability_assessment_and_exploitation
offensive_security_methods
Cryptographic Foundations and Algorithms
4 ECTS
cryptography_foundations
symmetric_cryptography
public_key_cryptography
cryptographic_hashes
Cryptographic Protocols, Trust, and Security
5 ECTS
cryptographic_protocols
cryptographic_key_management
cryptographic_trust_infrastructure
digital_signatures
formal_protocol_security
Operating Systems Foundations and Services
4 ECTS
operating_system_foundations
operating_system_filesystems
system_calls
shell_scripting
Operating Systems Processes and Resource Management
5 ECTS
processes_threads
concurrent_synchronization
cpu_scheduling
memory_management
io_management
Virtualized Systems Administration Foundations
5 ECTS
user_level_operating_systems
linux_network_configuration
linux_process_and_resource_monitoring
unix_accounts_and_filesystem_authorization
system_logging_and_scheduled_automation
Virtualized Systems Administration Automation and Monitoring
4 ECTS
docker_resource_management_api
configuration_as_code
centralized_configuration_management
centralized_monitoring
Real-Time Systems Fundamentals
2 ECTS
real_time_foundations
real_time_scheduling
Advanced Real-Time Systems
3 ECTS
aperiodic_scheduling
priority_resource_access
real_time_operating_system_concepts
Human-Computer Interaction
6 ECTS
interaction_and_interface_foundations
user_experience_foundations
interaction_ideation_and_prototyping
user_research_methods
interactive_system_evaluation
usability_design_and_evaluation
Geospatial Foundations and Analysis
4 ECTS
gis_fundamentals
cartographic_representation
geospatial_data_management
geospatial_analysis
Context-Aware Systems and Applications
5 ECTS
context_aware_systems
location_aware_services
mobile_edge_service_management
supervised_activity_recognition
differential_drive_robot_applications
Concurrency Models and Formalisms
2 ECTS
concurrent_message_passing
concurrent_systems_formalisms
Concurrent Programming and Verification
2 ECTS
shared_memory_concurrency_programming
concurrent_program_model_checking
Computer Architecture Fundamentals
6 ECTS
computer_system_architecture
instruction_set_architecture
processor_organization_and_isa
memory_and_addressing_hardware
processor_io_and_interrupts
architecture_performance_foundations
Advanced Processor Execution and Performance
3 ECTS
computer_performance_measurement
pipelined_processor_design
advanced_instruction_execution
Digital Logic Fundamentals
5 ECTS
electronics_fundamentals_and_instruments
boolean_logic_basics
combinational_logic_design
sequential_logic_basics
synchronous_asynchronous_logic
Embedded Systems and Programming
4 ECTS
embedded_systems_foundations
embedded_bare_metal_programming
arduino_programming
serial_communication_protocols
Digital Systems and Cryptographic Arithmetic
6 ECTS
language_syntax_grammars
binary_number_operations
programmable_logic_families
chip_hardware_context
medical_artificial_intelligence
medical_3d_modeling_pipeline
Computer Networks Foundations
4 ECTS
computer_network_fundamentals
networking_foundations
network_architecture_and_layers
link_layer_and_lan_access
Internetworking, Transport, and Network Programming
5 ECTS
internetworking_and_ip
ip_network_operations
tcp_udp_and_congestion
transport_services
network_programming
Big Data Storage and Processing
6 ECTS
big_data_processing_modes
hadoop_big_data
parquet_columnar_storage
cloud_big_data_platforms
web_data_scraping
server_side_php
Network Communication Foundations
3 ECTS
physical_layer_transmission
network_switching
channel_reliability
Internet Services and Protocol Analysis
3 ECTS
application_layer_protocols
dns_and_name_systems
network_protocol_analysis
Wireless Networking and Mobility
3 ECTS
wireless_radio_systems
wireless_networking_technologies
transport_and_network_service_quality
Distributed Systems Foundations and Programming
4 ECTS
distributed_systems_foundations
distributed_communication
remote_procedure_invocation
message_oriented_middleware
Distributed Algorithms, Consistency, and Coordination
5 ECTS
distributed_algorithms
distributed_consistency_and_replication
distributed_consensus_and_ordering
distributed_coordination_models
distributed_system_scalability
Blockchain and Cryptocurrencies
6 ECTS
blockchain_and_cryptocurrency_foundations
blockchain_principles
blockchain_network_organization
blockchain_platforms
blockchain_operations_and_contracts
blockchain_security_and_anonymity
Web Programming Foundations
3 ECTS
internet_and_web_foundations
web_protocol_and_uri_foundations
web_markup_and_content_formats
Web Application Development
6 ECTS
client_side_web_programming
client_side_web_frameworks
event_driven_server_programming
server_side_web_programming
rest_web_apis
web_application_architecture_and_state
Mobile Application Development Foundations and Android
6 ECTS
mobile_application_types
mobile_application_design
mobile_architecture_patterns
android_application_foundations
android_classic_user_interfaces
android_data_management
iOS Application Development
3 ECTS
ios_application_foundations
ios_uikit_interaction
ios_swiftui
Web Standards and Cryptographic Arithmetic
6 ECTS
web_standards
cryptographic_modular_arithmetic
web_data_scraping
internet_applications
server_side_php
computer_instruction_execution
Cloud Computing Platforms
4 ECTS
cloud_computing_foundations
cloud_infrastructure_and_service_models
cloud_service_orchestration
container_virtualization
Big Data Processing
5 ECTS
big_data_context_and_management
big_data_processing_architectures
mapreduce_programming
spark_big_data_processing
online_stream_processing
Edge IoT and Web of Things
6 ECTS
esp32_idf_programming
web_of_things
edge_and_fog_computing
iot_streaming_with_kafka
iot_interoperability_and_constraints
smart_agriculture_iot
Network Analysis and Testing
6 ECTS
large_network_analysis
network_hypothesis_testing
network_analysis_foundations
network_level_metrics
network_node_metrics
network_visualization